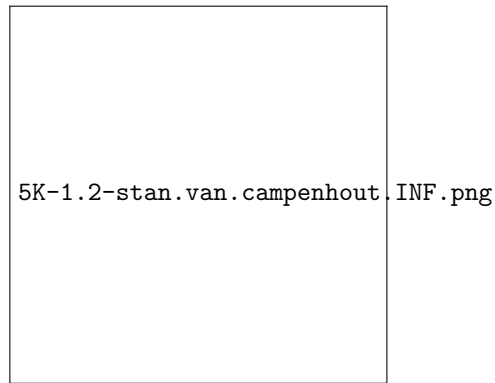


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$$r = \theta^2$$

$$r' = 2\theta \Rightarrow (r')^2 = 4\theta^2$$

$$r^2 = \theta^4$$

$$\begin{aligned} \Rightarrow \int_0^{4\sqrt{2}} \sqrt{\theta^4 + 4\theta^2} d\theta &= \int_0^{4\sqrt{2}} \theta \sqrt{\theta^2 + 4} d\theta = \frac{1}{2} \int_0^{4\sqrt{2}} \sqrt{\theta^2 + 4} d(\theta^2 + 4) \\ &= \left[\frac{(\theta^2 + 4)^{\frac{3}{2}}}{3} \right]_0^{4\sqrt{2}} = \frac{6^3}{3} - \frac{2^3}{3} = \frac{208}{3} \end{aligned}$$

References